Expt No: 8
Date: 20/3/26

NMAP to Discover Live Hosts Using Nmap Scans (ARP, ICMP, TCP/UDP) on the TryHackMe platform.

Aim:

To Study the working of the Nmap tool and discover live hosts in a network using different host discovery techniques such as ARP Scanning, ICMP Scanning, TCP Scanning and UDP Scanning on the TryHackMe platform. The experiment also aims to analyze the behaviour of different protocols and compare the result obtained from various scanning methods.

Steps:

1. Open the TryHackMe platform and Select the Nmap room
2. Study the network topology and identify the Source and destination hosts.
3. Use ARP requests to discover the MAC addresses of devices available in the Local Network.
4. Observe how ARP packets are broadcast within the network and identify the devices that receive the request.
5. Use the ICMP protocol to send Ping requests between two System and observe the responses.
6. Analysis the packet flow and identify the exchange of ARP packets before ICMP communication.
7. Perform ARP host discovery using the Nmap command and identify all active hosts in the Local network.

Output:

ARP :

Sud: nmap -PR -Sn 10.10.200.0/24

Staring Nmap

Nmap Scan report from 10.10.210.1

Host is up

Nmap done

ICMP.

Nmap -PE - Sn 10.10.210.0/24

Starting Nmap

Nmap Scan report for 10.10.210.6

Host is up

Nmap.done

TCP:

nmap - ps22 - Sn 10.10.210.0/24

Sudo nmap PA 80. Sn 10.10.210/24

8. Perform ICMP Echo Scanning and observe the hosts that respond to the Scan.
9. Perform ICMP Timestamp Scanning and analyze the response generated by the target System.
10. Perform ICMP Address Mask Scanning and Compare the result with the previous Scans.
11. Perform TCP SYN Scanning to identify live hosts by analysing SYN-ACK packets
12. Perform TCP ACK Scanning to verify the accessibility of the target ho
13. Perform UDP Scanning and observe how Nmap identifies active hosts through UDP packets.
14. Perform a Reverse DNS lookup to obtain the hostnames of the discovered Systems.
15. Compare the result obtained from ARP, ICMP, TCP and UDP Scanning techniques.
16. Analyze the advantages and limitations of each Scanning method.

Result: Thus, live hosts in the network were Successfully identified using different Nmap Scanning techniques. ARP Scanning was found to be effective for discovering devices in the Local network, while ICMP, TCP and UDP Scanning techniques were useful for identifying active hosts and verifying network connectivity. The experiment also demonstrated the effectiveness of the Nmap tool in network discovery and analysis.